PREPARATION AND CHARACTERIZATION OF MONO AND BIMETALLIC HZSM-5 CATALYSTS FOR THE CATALYTIC UPCYCLING OF USED COOKING OIL TO AROMATICS

A Summer Research Internship Report

Submitted by
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Duration: 2 Months

Declaration

I, Sahil Thakur, student of B.Tech in Chemical Engineering at IIT (ISM) Dhanbad, hereby declare that this internship report entitled "UCO to Aromatics conversion" is a genuine record of work carried out by me at the Lab, IIT Ropar, under the supervision of Dr. Navin Gopinathan. This report has not been previously submitted to any institution for any academic qualification or certificate.

I further declare that all the experimental work, image analysis, simulation tasks, and written content presented in this report are my own, and due credit has been given wherever reference has been made to the work of others.

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1. Introduction

Used cooking oil (UCO) is generated in large volumes by households, restaurants, and food processing units, and its disposal remains a persistent environmental and economic problem. When discharged into drains or landfills, UCO contaminates water bodies and soil, while its accumulation represents a wasted feedstock of long-chain hydrocarbons that could otherwise be converted into industrially valuable products [WCOReview2024]. Catalytic pyrolysis has emerged as one of the most promising routes for valorizing UCO, converting it into aromatic hydrocarbons such as benzene, toluene, and xylenes (BTX) that serve as building blocks for the petrochemical industry [Energies2022]. The efficiency and selectivity of this conversion depend strongly on the catalyst used, which makes catalyst design and synthesis a central problem in UCO valorization research.

Among the catalytic materials investigated for this purpose, zeolite based catalysts such as HZSM-5 have attracted particular attention because of their high surface area, well defined microporous structure, thermal stability, and strong Brønsted acidity. These properties make HZSM-5 especially effective for reactions that require shape selective aromatization, including biomass and waste oil pyrolysis, hydrocarbon reforming, and carbon dioxide utilization [Rasouli2022]. Reviews of waste oil valorization have reported that HZSM-5-catalyzed pyrolysis of used cooking oil can convert more than half of the feedstock into aromatic compounds, underscoring the potential of this catalyst family for the present work [Energies2022].

The catalytic performance of HZSM-5 can be further improved by loading it with transition metals such as zinc, copper, cobalt, chromium, and molybdenum. Metal loading modifies the acidity, redox behaviour, and active-site density of the catalyst, which in turn improves activity and product selectivity. Bimetallic catalysts often outperform their monometallic counterparts because the interaction between two different metals creates synergistic effects that are absent in single-metal systems; for instance, a Zn-Mo/HZSM-5 catalyst has been reported to raise BTEX yields to nearly 20 wt% by increasing the total number of acid sites available for aromatization [Moogi2022]. This synergy motivated the choice of zinc as a common element paired separately with copper, cobalt, and chromium in the bimetallic catalysts prepared during the internship, as discussed further in the Literature Review.

The work reported here was carried out during a Summer Research Internship at the Indian Institute of Technology Ropar, in the field of heterogeneous catalysis, under the guidance of the project supervisor. The internship focused on the preparation of metal loaded HZSM-5 catalysts by the wet impregnation method, the synthesis of bimetallic catalysts, pellet formation for catalytic testing, operation of a fixed bed reactor, BET surface area analysis, and the synthesis of ceria nanostructures. Alongside the experimental work, the theoretical background of catalyst preparation, characterization, and structure activity relationships was studied through peer reviewed literature accessed via Google Scholar, Consensus, and Elicit.

This report is organized to follow the progression of the internship. It begins with a review of the literature that guided the choice of HZSM-5 as the support and of zinc, copper, cobalt, and

chromium as the loaded metals. It then describes the synthesis of the mono and bimetallic catalysts, the physical observations recorded during preparation, and the subsequent pellet formation and fixed bed reactor setup used for catalytic testing. The report closes with a summary of the outcomes achieved and the technical skills developed during the training period.

2. Literature Review

Before the experimental work began, the literature on HZSM-5-based catalysts for biomass and waste-oil conversion was reviewed to identify suitable catalyst compositions and preparation routes. One study of particular relevance examined the catalytic pyrolysis of food waste using zinc loaded, molybdenum loaded, and zinc molybdenum bimetallic HZSM-5 catalysts under different gas environments [Moogi2022]. Introducing a hydrogen rich atmosphere significantly improved the yield of aromatic hydrocarbons such as benzene, toluene, ethylbenzene, and xylenes (BTEX) by promoting hydrodeoxygenation and hydropyrolysis while suppressing coke formation on the catalyst surface. Among all catalysts tested, Zn-Mo/HZSM-5 gave the highest BTEX yield, close to 20 wt%, because the combined effect of zinc and molybdenum increased the total number of acid sites available for aromatization, hydrodeoxygenation, and hydrodealkylation reactions [Moogi2022]. These catalysts were prepared by incipient wetness impregnation: commercial HZSM-5 was loaded with zinc nitrate and ammonium molybdate individually or together, then dried at 110°C for 12 hours and calcined at 550°C for 3 hours [Moogi2022].

The same study characterized its catalysts using nitrogen physisorption and ammonia temperature-programmed desorption (NH_3-TPD) to quantify surface area, pore volume, and acid site density, since these properties directly influence aromatization and hydrodeoxygenation activity [Moogi2022]. This reinforced the decision to prioritize acidity and surface area when selecting a support and dopant combination for the present work.

HZSM-5 was selected as the support for this work, rather than other zeolites such as HY or beta-zeolite, because of its narrow pore aperture, which restricts the formation of bulky polyaromatic coke precursors and favours the shape-selective formation of BTX aromatics, together with its strong Brønsted acidity, which is essential for the cracking, dehydrogenation, and aromatization steps of oil pyrolysis [Rasouli2022]. Zinc was chosen as the primary dopant because it is consistently reported to enhance aromatization activity in HZSM-5 by generating $ZnOH^+$ or ZnO species that promote dehydrogenation of pyrolysis intermediates [Pan2021] [Niu2014]. Copper, cobalt, and chromium were then selected as co-dopants to compare their effect on redox behaviour and acid-site distribution when combined with zinc, following the precedent set by bimetallic systems such as Zn-Mo/HZSM-5 and Ni-Zn/ZSM-5 reported in the literature [Barzallo2023] [Moogi2022]. This comparison was intended to identify which metal combination gave the most favourable balance of acidity and dispersion among the bimetallic catalysts synthesized during the internship.

The review also covered alternative preparation routes to wet impregnation, including wet ion exchange of pre formed HZSM-5 with mixed Zn-Co or Zn-Mo solutions [Aziz2018] [Vichaphund2013] [Li2023], in-situ hydrothermal incorporation of zinc during zeolite crystallization followed by sequential ion exchange of a second metal [Pan2021], and supercritical solvothermal deposition, reported to give unusually high atomic-level metal dispersion compared with incipient impregnation [Julian2020]. These routes generally achieve better metal dispersion and stronger metal-support interaction than wet impregnation, but require specialized equipment or longer processing times. Reviewing them clarified why wet impregnation, despite its comparatively lower dispersion, remained the appropriate choice for the mono and bimetallic catalysts synthesized within the scope and timeline of this internship.

3. Materials and Catalyst Synthesis

The primary objective of this stage of the internship was to prepare mono and bimetallic HZSM-5 catalysts for subsequent testing in a fixed bed reactor. All catalysts were prepared by wet impregnation, chosen for its simplicity, reproducibility, and ability to achieve reasonably uniform metal distribution on porous supports, consistent with the literature reviewed above.

Four monometallic catalysts were first prepared by loading zinc, copper, cobalt, and chromium separately onto HZSM-5 using their respective metal nitrate precursors, so that the properties of each metal could be compared individually before combining them. For each catalyst, 2 g of HZSM-5 was dispersed in 20 mL of deionized water to form a uniform suspension, while the required quantity of metal nitrate hydrate — calculated to give a 5 wt.% metal loading — was dissolved separately in another 20 mL of deionized water.



Figure 1. General view of the impregnation setup used for preparing the catalyst suspensions.

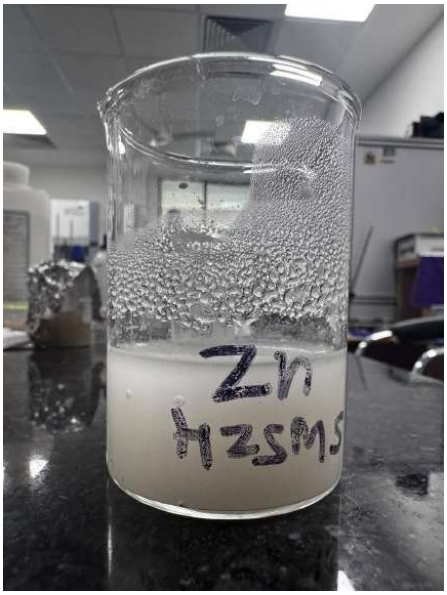


Figure 2. Zn/HZSM-5 suspension.



Figure 3. Co/HZSM-5 suspension.



Figure 4. Cr/HZSM-5 suspension.

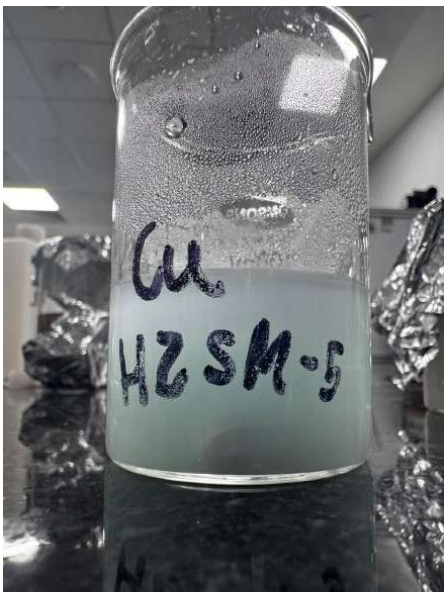


Figure 5. Cu/HZSM-5 suspension.

The metal precursor solution was added slowly to the HZSM-5 suspension under continuous stirring for four hours at room temperature to distribute the metal ions evenly over the support (Figure 6).

The same procedure was then extended to prepare three bimetallic catalysts — Zn-Cu/HZSM-5, Zn-Co/HZSM-5, and Zn-Cr/HZSM-5 — in which solutions containing two metal precursors were prepared and impregnated onto HZSM-5 simultaneously, so that the resulting catalysts could combine the properties of both metals and potentially benefit from synergistic interactions between the active species, following the same rationale established in the Literature Review [Barzallo2023].



Figure 6. Zn-Cr bimetallic precursor. | *Figure 7. Zn-Co bimetallic precursor.*

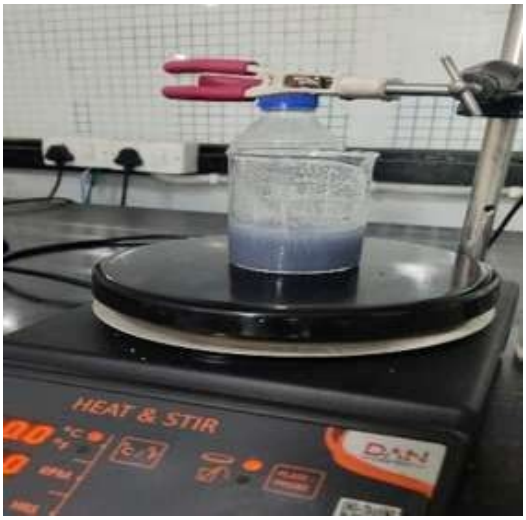


Figure 8. Zn-Cu bimetallic precursor.

After stirring, each catalyst suspension was centrifuged twice for twenty minutes to separate the solid catalyst from the liquid phase and remove unadsorbed metal ions. The recovered catalyst was dried overnight at 80°C, then calcined at 500°C for six hours in a muffle furnace, during which the metal nitrate precursors decomposed into their corresponding oxides and became fixed on the HZSM-5 support. The calcined catalysts were ground into fine powders, labelled as Zn/HZSM-5, Cu/HZSM-5, Co/HZSM-5, and Cr/HZSM-5 (with pure HZSM-5 retained as a reference), and stored in labelled vials pending further use.

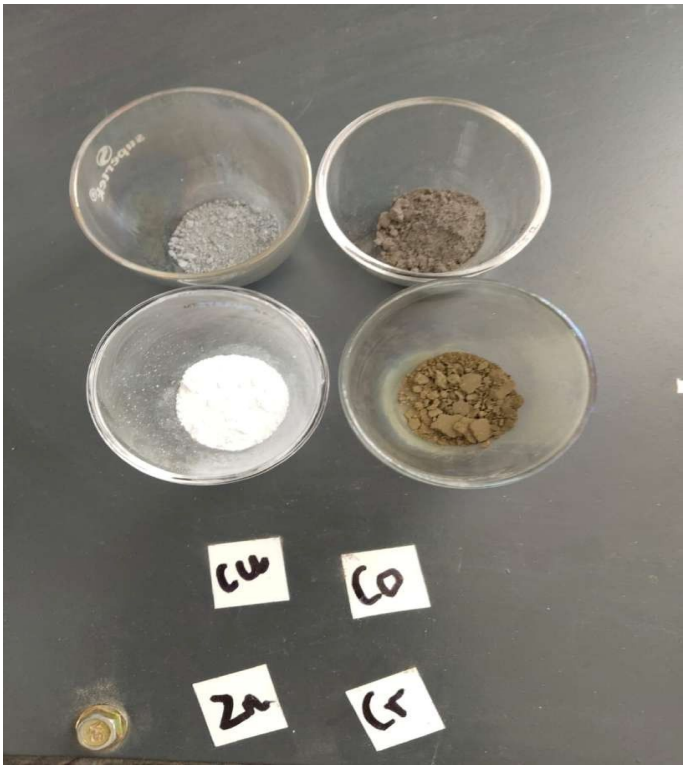


Figure 9. Catalyst samples recovered after calcination, labelled Cu, Co, Zn, and Cr.

Catalyst	Mass recovered after calcination (mg)
Zn/HZSM-5	2054.5
Cr/HZSM-5	3528
Cu/HZSM-5	1979
Co/HZSM-5	1981



Figure 10. Calcined mono-metallic catalyst powders prior to grinding.

Throughout the synthesis, careful attention was paid to maintaining consistent stirring, accurate weighing, and controlled heating during drying and calcination, since these parameters influence metal dispersion, crystallinity, surface area, and ultimately catalytic activity.

4. Observations During Synthesis

During drying, distinct colour differences were observed among the catalysts, arising from the hydrated transition metal nitrate precursors deposited on the zeolite surface: Cu/HZSM-5 appeared light blue, Co/HZSM-5 turned pink, Cr/HZSM-5 showed a dark green colour, and Zn/HZSM-5 remained cloudy white.

Cu/HZSM-5 and Cr/HZSM-5 adhered strongly to the glass beakers after drying, in some cases causing breakage during scraping. This was resolved by rewetting the deposited catalyst with a small quantity of deionized water and returning the beakers to the drying oven at 80°C overnight, which substantially improved recovery.

Calcination produced further colour changes: Cu/HZSM-5 turned from light blue to grey, Co/HZSM-5 from pink to black, and Cr/HZSM-5 to a darker faded green, while Zn/HZSM-5 remained white. These changes reflect the removal of moisture and nitrate groups and the conversion of the metal precursors into stable oxide phases; heat treatment of this kind is known to alter the oxidation state and crystallinity of supported metal species, which in turn affects their optical and catalytic properties [Barzallo2023].

The catalysts were successfully recovered, ground into a uniform particle size, and stored in labelled vials for pelletization and further characterization.

5. Pellet Formation

Because powdered catalysts cannot be packed directly into a fixed bed reactor, both the monometallic and bimetallic catalysts were compressed into pellets using a hydraulic press after synthesis and calcination. The pellets were then crushed and passed through a standard BSS-14 sieve to obtain particles of uniform size, which is important for proper packing of the catalyst bed, minimizing pressure drop, and ensuring uniform gas flow during reactor operation.

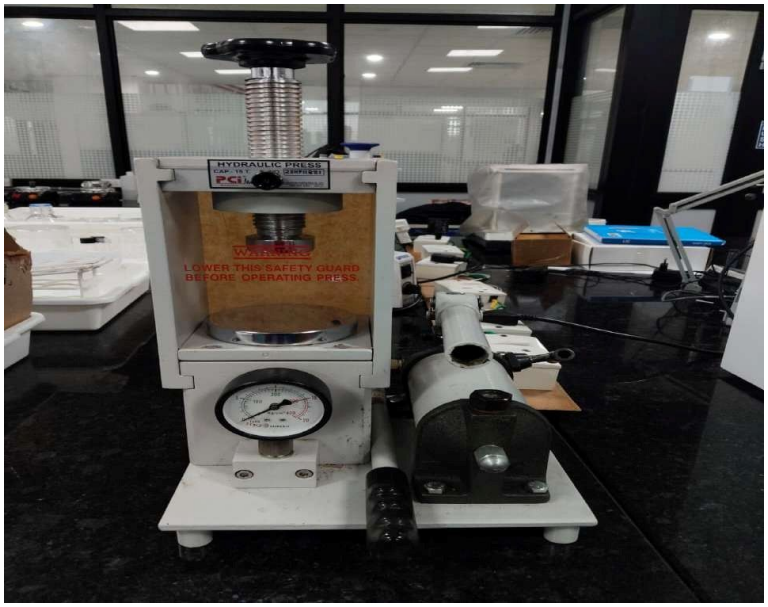


Figure 11. Hydraulic press used for compressing catalyst powder into pellets.

For each catalyst, the powder was placed in the die of the hydraulic press and compressed under high pressure to form a hard pellet, which was then broken down and sieved; only particles retained within the required BSS-14 size range were collected for reactor studies. This procedure was applied identically to all seven catalysts.

Monometallic catalyst	BSS-14 mass (mg)	Bimetallic catalyst	BSS-14 mass (mg)
Zn/HZSM-5	1796	Zn-Cu/HZSM-5	1118.3
Cu/HZSM-5	1943	Zn-Cr/HZSM-5	1042.5
Co/HZSM-5	1673	Zn-Co/HZSM-5	1213.6
Cr/HZSM-5	1683		

For every catalyst, more than 1.5 g of BSS-14 particles was obtained, which was sufficient to pack the catalyst bed of the fixed bed reactor. The prepared particles were stored in labelled vials, ready for catalytic testing.

6. Fixed Bed Reactor Setup



Figure 12. Laboratory-scale fixed bed reactor used for catalytic testing.

The fixed bed reactor was one of the most widely used systems for studying heterogeneous catalytic reactions, since the catalyst remained stationary while the reactant gases flowed continuously through it. During the internship, the working principle and operation of the laboratory-scale fixed bed reactor used for catalytic testing of the synthesized catalysts were studied in detail. The reactor consisted of a quartz reaction tube placed inside an electrically heated tubular furnace. Before each experiment, the catalyst pellets were loaded inside the quartz tube between two plugs of quartz wool, which held the catalyst bed in position while allowing gases to flow uniformly through it (Figures 13 and 14). Once loaded, the furnace was heated to the desired reaction temperature, typically between 300°C and 500°C depending on the reaction under study, and the temperature was monitored until it stabilized. Reactant gases were supplied from high pressure cylinders, with individual flow rates controlled using Mass Flow Controllers (MFCs) to ensure accurate control of gas composition and total flow. After the required temperature and flow conditions were reached, the reactant gases passed through the catalyst bed, where the desired catalytic reactions took place, producing methanol, carbon monoxide, methane, and other hydrocarbons depending on the catalyst and operating conditions. The hot gaseous products leaving the reactor were passed through a condenser connected to a chiller, which cooled the stream and condensed liquid products such as methanol and water while non-condensable gases remained in the gas phase. The condensed liquid was collected in a receiving bottle for further analysis, while the remaining gases were directed to a Gas Chromatograph (GC), which provided information on reactant conversion, product distribution, and catalyst performance.





Figure 13. Catalyst bed packed between quartz wool plugs inside the reactor tube. | Figure 14. Catalyst bed loaded into the fixed bed reactor.

A typical reactor run proceeded in four stages. First, the equipment was switched on and the reactor feed was loaded into the system. The furnace temperature was then raised to 500°C over two hours to activate the catalyst, held at that peak for five minutes, and lowered to the reaction temperature of 480°C over the next twenty minutes. The reaction was then run at 480°C for five to six hours to allow the feed to react fully, with a volumetric flow rate of 1 mL/hr maintained throughout; the nitrogen carrier flow was set to 5 mL/hr when the feed was off and increased to 10 mL/hr once the feed was on. The first product drop, taken as the reference time T1, was typically observed around 1.5 hours into the run, after which gas samples were collected at T1+60, T1+120, T1+180, and T1+240 minutes to track the progress and direction of the reaction.



Figure 15. Magnetic stirrer hotplate used during impregnation.



Figure 16. Additional stirring and heating stations used to process multiple batches in parallel.



Figure 17. Ultrasonic cleaner used for sample dispersion.



Figure 18. Centrifuge station used for solid-liquid separation.



Figure 19. Glassware oven



Figure 20. Oven for drying of solutions



Figure 21. Third Oven





Figure 22. Standard BSS-14 and BSS-18 test sieves used for pellet grading. *Figure 23. Reagent-grade metal precursor stock used for catalyst synthesis.*



Figure 24. Catalyst bed viewed inside the reactor quartz tube before a run.

7. Conclusion

This internship provided practical exposure to the complete workflow of heterogeneous catalyst development, from support and metal selection through synthesis, pelletization, and reactor loading. Four monometallic catalysts (Zn/HZSM-5, Cu/HZSM-5, Co/HZSM-5, and Cr/HZSM-5) and three bimetallic catalysts (Zn-Cu/HZSM-5, Zn-Co/HZSM-5, and Zn-Cr/HZSM-5) were successfully prepared by wet impregnation, characterized qualitatively through the colour

changes observed during drying and calcination, and pelletized to a uniform BSS-14 particle size suitable for packing a fixed bed reactor.

The literature review clarified why HZSM-5 was an appropriate support for aromatization reactions and why zinc-based bimetallic combinations were prioritized over monometallic catalysts, based on evidence that bimetallic acid-site synergy improves BTEX selectivity in comparable pyrolysis systems. Alongside the synthesis work, the internship built familiarity with the operating principles of a fixed bed reactor, mass flow control, gas chromatography-based product analysis, and BET surface area characterization, together with practical skills in handling laboratory equipment such as hydraulic presses, muffle furnaces, and centrifuges.

Overall, the internship strengthened the technical foundation needed for catalyst-focused research and confirmed the potential of zinc-based bimetallic HZSM-5 catalysts for the pyrolytic upcycling of used cooking oil into value-added aromatics, a direction that could be extended through catalytic testing of the pelletized samples prepared here (Figures 25 and 26).



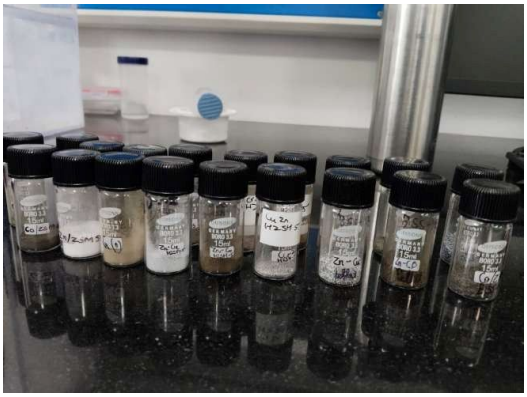


Figure 25 & 26. Complete set of monometallic and bimetallic catalyst samples prepared during the internship.

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